

RAHUL RAMESH OMALUR

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Professional Summary

Applied AI Engineer with a Forward Development Engineering (FDE) mindset, building production-grade AI and web applications across **LLMs, agentic AI, machine learning, backend APIs, data pipelines, cloud deployment, and full-stack product integration**. Strong fit for **AI Engineer, Machine Learning Engineer, LLM Engineer, Data Scientist, Data Analyst, and Full-Stack Developer** roles.

Education

IIIT Delhi (Tier-1 Research-Led Institute)

Nov 2022 – May 2026

Bachelor of Technology with **Honors** in Computer Science and Artificial Intelligence (**CGPA: 8.69**)

Relevant Coursework: Machine Learning, Deep Learning, LLMs, NLP, Computer Vision, MLOps, DBMS, Operating Systems

Experience

TCS Research & Innovation

May 2025 – July 2025

Machine Learning Research Intern

- Engineered a **92-feature pipeline** (50 metadata + 42 ECG-derived features) over **318 patient records**, applying **PCA** and imbalance handling on a skewed CAD-to-normal ratio (**197:22**).
- Benchmarked **4 classifiers** across **19 experiments**; best RFC (ECG+Meta) achieved **0.87 accuracy** and **0.93 F1**, with HistGBC attaining highest minority-class sensitivity (**G-score: 0.61**).
- Surfaced a clinical trade-off between overall accuracy and CAD-recall (F1-flip up to **0.38** vs. near-zero for SVM/RFC), informing model selection for high false-negative-risk deployment.

CodePlex Technologies Private Ltd.

May 2024 – July 2024

Software Engineer Intern

- Worked on a **PHP Laravel web application** to automate scraping and processing of user-submitted obituary data, integrating **2–3 vendor-specific source websites daily** into a unified pipeline.
- Implemented vendor-specific web scrapers using **Simple HTML DOM Parser**, handling site-by-site structural variation to migrate legacy datasets into structured **MySQL repositories**.
- Diagnosed and resolved **data pipeline anomalies** to maintain schema consistency and accuracy across continuously ingested cross-platform records.

Projects

SmartReach AI – Email Campaign Manager | *Django, Next.js, LLMs, scikit-learn*

Jan 2025 – May 2025

- Built an end-to-end email marketing platform with **LLM-powered content generation, send-time optimization**, and real-time analytics across the campaign lifecycle.
- Architected a decoupled **Django REST + Next.js** system with JWT auth, contact segmentation, campaign scheduling, and create-send-track workflows.
- Integrated **DPDP Act (2023)**-aligned consent and unsubscribe flows for compliant regulated marketing use cases.

SentinelAI – Intrusion Detection System | *Flask, MongoDB, OpenCV, PyTorch, Next.js*

Jan 2026 – May 2026

- Built a **Flask REST API** backend with **Redis** face-embedding cache and **MongoDB** identity storage for a real-time threat-scoring pipeline across a **Next.js** dashboard.
- Developed a CV inference pipeline with face recognition, ByteTrack tracking, action recognition, and **ONNX Runtime** quantization on Raspberry Pi 4, cutting memory by **40–55%**.
- Reduced known-subject latency from **51.7ms** to **23.9ms** via Redis caching, achieving **85.06% identity accuracy**.

Nexus Agent – AI Network Simulation Analyst | *LangGraph, Next.js, Flask, SQLite, MCP*

Jan 2026 – May 2026

- Developed a **Flask REST API + Next.js** platform with workspace-isolated **SQLite** for multi-tenant natural-language querying over AstraSim simulation logs.
- Created schema-aware **Text-to-SQL** with execution-guided self-correction via **Groq API**, achieving **90–100% SQL accuracy** on a **1,000-run** dataset.
- Designed a **LangGraph** multi-agent system with SQL, visualization, and insight agents plus sandboxed Matplotlib plot generation.

Technical Skills

Languages: Python, SQL, Java, C, PHP, JavaScript

Frameworks & Libraries: Hugging Face, FastAPI, Flask, Django, Laravel, PostgreSQL, MongoDB, PyTorch, TensorFlow, LangChain, MCP, DSPy

Cloud & Tools: AWS, GCP, Groq, Git, Docker, Kafka, Spark, n8n, Redis, Celery, Simple HTML DOM Parser

Leadership / Extracurricular

AudioBytes Coordinator – Organized technical/musical events and led the college band in inter-college competitions.